Evaluation Report & Next-Step Execution Roadmap

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Assigned Construct: Fluid Intelligence (Gf)

Proposed Task: Rule Discovery Task

Evaluation Status: Proceed with Modifications and Enhanced Adaptive Reasoning Framework

1. Executive Summary

The proposed Rule Discovery Task demonstrates exceptional alignment with our vision of developing an AI-driven cognitive assessment platform that measures higher-order reasoning abilities through engaging, adaptive, and culturally fair experiences. The task requires students to infer hidden rules from novel symbols and apply these rules to unfamiliar situations without relying on prior knowledge. The proposed framework directly measures core dimensions of Fluid Intelligence, including inductive reasoning, abstract reasoning, pattern recognition, deductive reasoning, and logical problem solving. However, if the rule structures become excessively complex or involve remembering too many symbols simultaneously, the task may unintentionally measure Working Memory and Processing Speed. To align with our startup objectives, the assessment should prioritize reasoning quality, rule extraction, and transfer of learning while controlling memory demands and unnecessary time pressure.

2. Alignment with MentiScope's Objectives

• Alignment with CHC-Gf Theory: 9.5/10

• Scientific Validity: 9.5/10

• Innovation: 9.5/10

• Adaptive Potential: 9.5/10

• AI Readiness: 9.5/10

• MVP Feasibility: 9/10

• Overall Evaluation: 9.5/10 – Proceed with Modifications

3. Scientific Assessment

Primary Fluid Intelligence components to measure:
• Inductive Reasoning
• Deductive Reasoning
• Abstract Reasoning
• Pattern Recognition
• Logical Problem Solving

Potential construct contamination:
• Working Memory (Gsm)
• Processing Speed (Gs)
• Test-taking strategies
• Visual complexity effects

The task should primarily assess novel reasoning and rule discovery while minimizing memory load and speed dependence.

4. Recommended Task Architecture

Module 1 – Pattern Observation
Students observe examples of unfamiliar symbols and relationships.
Measures: Pattern recognition and attention to relationships.

Module 2 – Rule Discovery
Students infer hidden rules through inductive reasoning.
Measures: Inductive reasoning and abstract thinking.

Module 3 – Rule Application
Students apply discovered rules to new examples.
Measures: Deductive reasoning and transfer of learning.

Module 4 – Multi-Step Challenge
Students solve increasingly complex reasoning puzzles with adaptive difficulty.
Measures: Logical problem solving and reasoning flexibility.

5. Recommended Scoring Model

• Pattern Recognition Score – 20%

• Rule Discovery Score – 30%

• Rule Application Score – 25%

• Logical Problem Solving Score – 25%

6. AI Analytics to Capture

• Rule discovery time

• Accuracy of inferred rules

• Transfer accuracy to new cases

• Number of hypothesis revisions

• Error patterns

• Difficulty progression

• Strategy shifts

• Reasoning efficiency profile

• Learning curve

• Persistence and exploration behavior

7. MVP Deliverables (Current Phase)

✓ Procedural symbol generator

✓ Rule generation engine

✓ Adaptive difficulty framework

✓ Minimum 5,000 unique symbol-rule combinations

✓ Behavioral scoring engine

✓ Event logger

✓ Analytics dashboard

✓ Documentation

✓ Pilot dataset

8. Assessment SDK Compliance Requirements

All outputs must follow the common Assessment SDK specification.

Session Data:
student_id, session_id, construct, task_id, module_version, difficulty_level, start_time, end_time, timestamp.

Event Data:
event_id, student_id, session_id, construct, task_id, item_id, event_type, response, correct, reaction_time_ms, error_type, difficulty_level, timestamp.

Score Output:
raw_score, normalized_score, percentile, sub_scores, confidence_score.

Analytics Output:
behavioral_metrics and recommendations.

9. Next Sprint Execution Plan (2 Weeks)

Sprint Objective:

Transform the Rule Discovery Task into a scientifically robust adaptive reasoning engine and develop the foundational MVP prototype.

Expected Deliverables:

1. Literature review on Fluid Intelligence (Gf) and CHC theory.
2. Construct map defining inductive, deductive, abstract, and logical reasoning.
3. Symbol and rule generation specifications.
4. Adaptive difficulty framework.
5. Behavioral scoring engine design.
6. Event logging implementation.
7. Analytics dashboard design.
8. Demonstration prototype of one adaptive reasoning module.

10. Definition of Done (DoD)

Scientific:

✓ Fluid Intelligence components clearly defined.

✓ Working Memory demands minimized.

✓ Rule complexity calibrated.

Technical:

✓ Procedural symbol generator implemented.

✓ Rule generation engine functional.

✓ Minimum 5,000 unique symbol-rule combinations generated.

✓ Event logging and analytics dashboard available.

✓ Results exported in SDK format.

Validation:

✓ Internal consistency ≥ 0.80.

✓ Test-retest reliability ≥ 0.80.

✓ Pilot study with at least 100 students.

✓ Correlation studies with established Gf assessments such as Raven's Progressive Matrices and CFIT.

Guidance to the Intern

Your responsibility is not to build the entire platform at this stage. Your responsibility is to build a scientifically valid micro-assessment engine for Fluid Intelligence that follows the common Assessment SDK. Focus on creating adaptive rule-discovery tasks, procedural content generation, and rich reasoning analytics. This module will later be integrated into the unified cognitive and career assessment platform.